

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
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Experiment 9
Clustering Human Activity Recognition Data using
K-Means, DBSCAN, and Hierarchical Clustering

1. Objective

- To implement and analyze unsupervised clustering algorithms on the Human Activity Recognition dataset.
- To apply and compare K-Means, DBSCAN, and Hierarchical Agglomerative Clustering.
- To visualize clusters using dimensionality reduction techniques.
- To evaluate clustering quality using both internal and external metrics.

2. Dataset

Dataset Used: Human Activity Recognition Using Smartphones Dataset

- Total volunteers: 30
- Activities: WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING
- Sensors used: 3-axis accelerometer and gyroscope sampled at 50 Hz
- Features: Time-domain and frequency-domain extracted features

3. Theory

3.1 K-Means Clustering

K-Means partitions the data into k clusters by minimizing within-cluster variance. Each point is assigned to the nearest centroid, and centroids are iteratively updated until convergence.

Algorithm Steps:

1. Initialize k centroids.
2. Assign each point to the nearest centroid.
3. Recompute centroids as the mean of assigned points.
4. Repeat until centroids stabilize.

3.2 DBSCAN

DBSCAN groups together points that are closely packed and marks points in low-density regions as noise. The key parameters are ε (radius) and `min.samples`.

3.3 Hierarchical Agglomerative Clustering

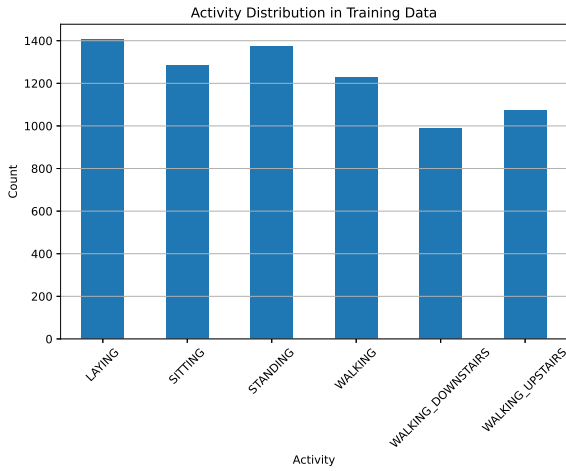
Hierarchical clustering builds a hierarchy of clusters by iteratively merging the closest clusters. Ward linkage is commonly used to minimize variance within clusters.

4. Implementation Details

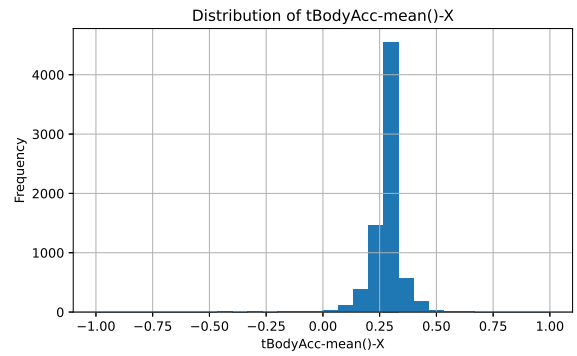
- Loaded the HAR training dataset from the extracted zip file.
- Assigned feature names using `features.txt`.
- Mapped numeric activity labels to activity names.
- Checked for missing values and verified dataset consistency.
- Standardized the feature matrix using z-score normalization.
- Applied PCA for 2D visualization and dimensionality reduction for clustering analysis.
- Tuned K-Means using Elbow Method and Silhouette Score.

- Tuned DBSCAN using different values of ε and `min_samples`.
- Applied Agglomerative Clustering and generated a dendrogram.
- Evaluated clustering results using Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index, Adjusted Rand Index, and Normalized Mutual Information.

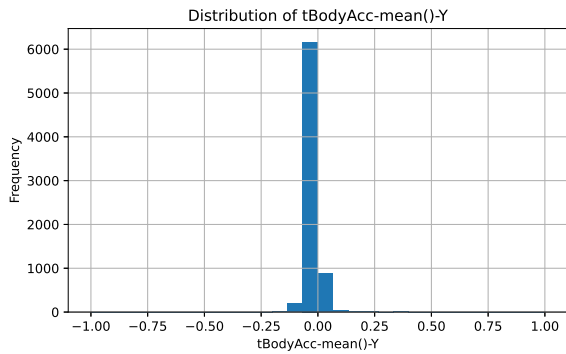
5. Visualizations



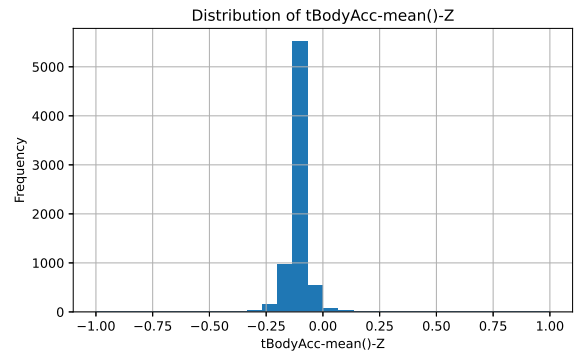
(a) Class distribution / activity distribution



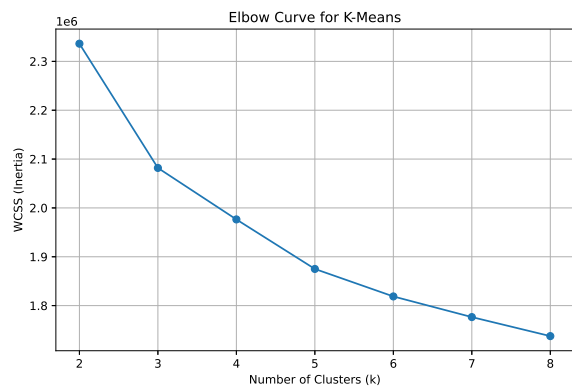
(b) Feature distribution / summary visualization



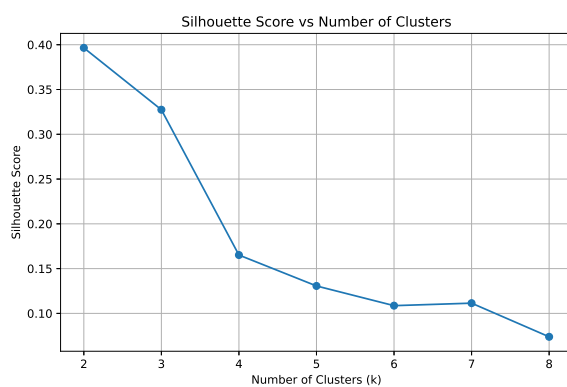
(a) Feature distribution



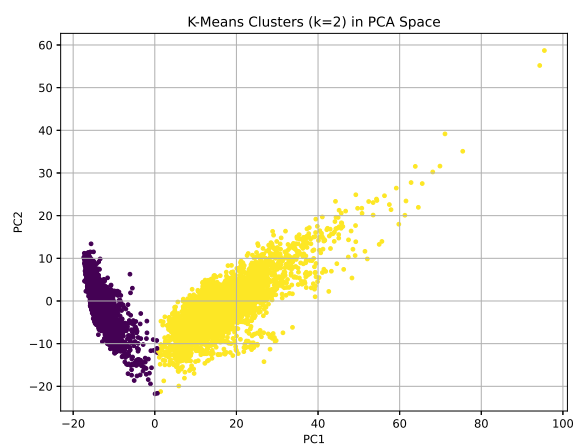
(b) Feature distribution / summary visualization



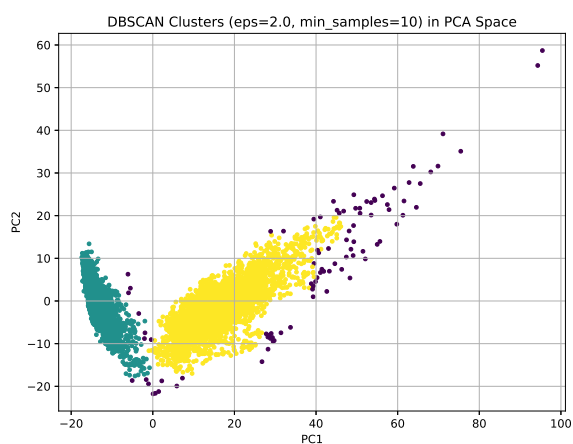
(a) Elbow curve: k vs WCSS



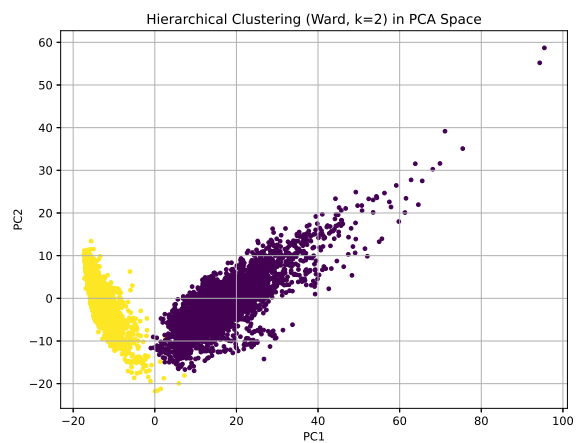
(b) Silhouette curve: k vs silhouette score



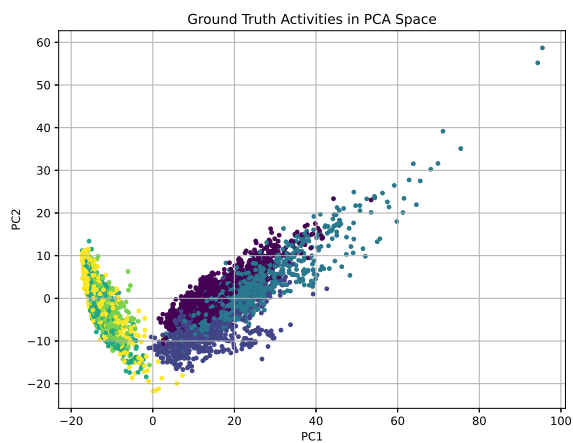
(a) PCA plot for K-Means clusters



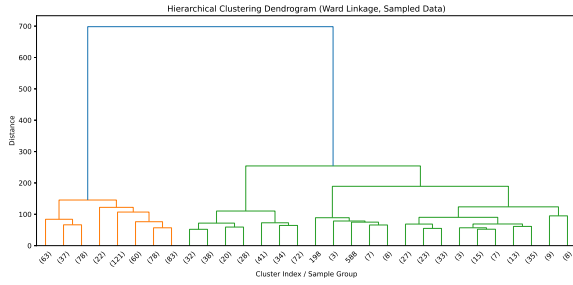
(b) PCA plot for DBSCAN clusters



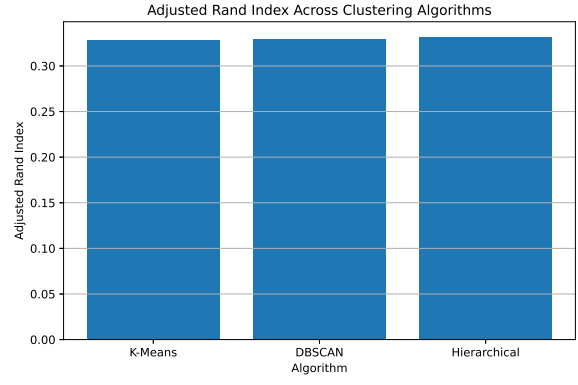
(a) PCA plot for Hierarchical clustering



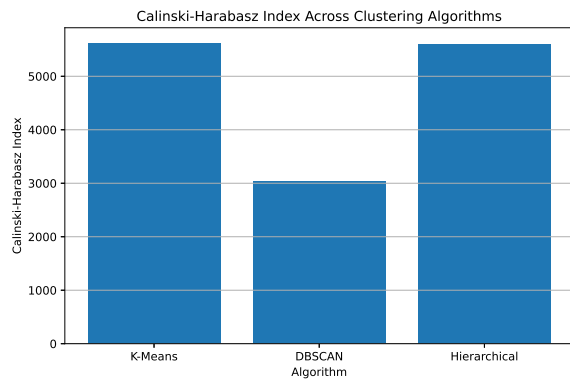
(b) Ground truth activity labels in 2D



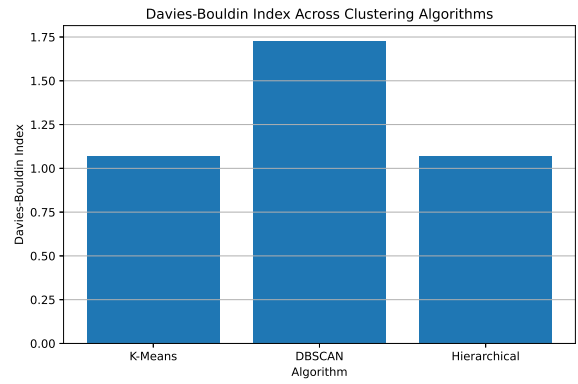
(a) Hierarchical clustering dendrogram



(b) Bar plot of evaluation metrics across algorithms



(a) Bar plot of evaluation metrics across algorithms



(b) Bar plot of evaluation metrics across algorithms

6. K-Means Elbow Method Results

Table 1: K-Means Elbow Method Results

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	2.336224e+06	0.3965
3	2.081e+06	0.3274
4	1.976e+06	0.1651
5	1.8750e+06	0.1306

Selected Best Value of k : 2

7. DBSCAN Tuning Results

Table 2: DBSCAN Parameter Evaluation

ε	min_samples	Clusters Found	Noise Points	Silhouette Score
2	10	2	91	0.7123
2	5	2	58	0.7098
1.5	3	6	54	0.5479
1.5	10	4	211	0.5239
1.2	10	5	320	0.4904

8. Hierarchical Clustering Results

Table 3: Hierarchical Clustering Summary

Linkage Method	Number of Clusters	Sillouhette Score
Ward	2	0.396056

9. Evaluation Metrics and Results

Table 4: Performance Comparison of Clustering Algorithms

Model	Silhouette	Davies–Bouldin	Calinski–Harabasz	ARI	NMI
K-Means	0.3965	1.0688	5626.01	0.3284	0.5432
DBSCAN	0.3925	1.7286	3029.62	0.3292	0.5388
Hierarchical	0.3960	1.0698	5608.27	0.3318	0.5554

10. Observation Questions and Answers

1. Which algorithm produced the most meaningful clusters? Why?

All Algorithms Produced The same amount and similar clusters, which is 2. The clusters can be divided into "stationary" or "Unstationary".

2. How sensitive was K-Means to the choice of k ?

the lower the k value, the greater the value of WCSS and thus error. it slowly decreases as k increases, though it may increase again due to overfitting.

3. Did DBSCAN detect noise or small clusters effectively?

Yes, it detected and identified the noise points.

4. **How does linkage choice affect hierarchical clustering?**

The choice of linkage in hierarchical clustering determines how distances between clusters are computed, thereby influencing the shape and quality of clusters. Single linkage considers the nearest points and often produces chain-like clusters with high sensitivity to noise. Complete and average linkage generate more compact and balanced clusters by considering farthest and mean distances respectively. Ward's linkage minimizes within-cluster variance, typically resulting in well-separated, spherical clusters and is generally preferred for structured datasets.

5. **Which internal metric best matched visual intuition of cluster quality?**

Hierarchical clustering performs best among the three models as it achieves the highest silhouette score and Calinski–Harabasz index while maintaining a low Davies–Bouldin index. It also shows the best agreement with true labels through higher ARI and NMI values, indicating superior clustering quality and structure compared to K-Means and DBSCAN.

11. Conclusion

Thus, i am able to identify the best performing clustering algorithm and read the internal and external metrics to determine as such, and tuned the hyperparameters to find the best fit of each algorithm.